

TORSIONVERSE SERIES 2: IMPLICATIONS AND APPLICATIONS

— SYNTHESIS

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TORSIONVERSE SERIES 2: IMPLICATIONS AND APPLICATIONS — SYNTHESIS

v3 adds Sections 4.2 expansion (Candidate D, RS14/RS15), Section 7 (spinning pyramid/escalator).

Companion script: analysis/demos/series2_doc.py (7/7 PASS, SD1-SD7)

ABSTRACT

Series 1 established the framework from first principles. Series 2 derives observable and testable implications across astrophysics, nuclear engineering, chemistry, cosmology, acoustics, human-scale geometry, and fluid mechanics. All series 2 results use ONLY series 1 parameters with zero additional free parameters. This document summarises the key results.

Reference papers (Series 1 DOIs in Section 7).

1. ASTROPHYSICAL PREDICTIONS

1.1 Black hole TDE jet delay: AT2018hyz (star eaten 2018, jet 2021)

OBSERVATION (Cendes et al. 2022, ApJ):

AT2018hyz: tidal disruption event, star eaten ~Oct 2018. No significant radio emission 2018-2020. In late 2021 (~3 years later): bright radio flare consistent with a mildly relativistic outflow ($v \sim 0.5c$). Standard physics: delayed jet launch. Torsionverse: jammed boundary shell reflection.

WHY STRAIGHT-THROUGH TRANSIT DOES NOT WORK:

For $M_{\text{BH}} \sim 5e7$ solar masses: $r_{\text{S}} = 1.48e11$ m. Diametric transit time = 0.3 hrs. For a 3-year transit: $r_{\text{BH}} = 1.5$ light-years, $M_{\text{BH}} = 5e12$ solar masses. AT2018hyz is nowhere near this mass. Interior-transit is ruled out numerically.

CORRECT TORSIONVERSE MECHANISM -- DISPLACING MAGNETIC FIELD + JAMMED SHELL:

The depleted zone (black hole interior) is created by a sustained T_{2g} (magnetic vortex) field from the rotating infalling matter sustained over the medium's relaxation timescale, draining cells from the interior. As cells drain they pile up at the boundary -- hyper-compressed, containing ALL cells formerly in the depleted volume. The boundary shell is the highest-density region achievable in the medium.

STELLAR DEBRIS HITS THE T_{2g} BOUNDARY (CG MECHANISM):

Infalling magnetized stellar debris carries its own T_{2g} (magnetic) field. At the boundary: $T_{2g}(\text{debris}) \times T_{2g}(\text{boundary}) = A_g + T_{2g} + H_g$ [CG exact]

A_g : further compresses boundary (more jamming, more depletion)

T_{2g} : radially reflected (matter bounced = accretion disk wind)

H_g : emitted perpendicular to impact plane = JET

$H_g = T_{1g} \times T_{2g} = \text{field strength tensor} = \text{synchrotron photons} + e^+/e^- \text{ pairs}$.

The jet IS H_g emission from the $T_{2g} \times T_{2g}$ boundary collision.

Jet direction: perpendicular to disk (normal to equatorial infall plane).

THE 3-YEAR DELAY IN TORSIONVERSE:

The disruption produces a stream of debris. For a jet to launch, the debris must first circularize into an accretion disk (viscous timescale). Only once the disk has formed does matter fall coherently onto the boundary shell at a sustained rate, building enough T_{2g} pressure for coherent H_g emission.

The 3-year delay = disk formation + pressure buildup timescale.
Standard viscous estimate ($\alpha=0.1$, $r_{\text{circ}}=100 r_S$): ~ 0.2 years [too fast].
Full disk formation with realistic stream-stream collision geometry: 1-3 years.

QUANTITATIVE PREDICTIONS (torsionverse, falsifiable):

- (a) Jet direction: perpendicular to accretion disk (same as GR prediction)
- (b) Jet speed: set by reflection angle off curved jammed shell $\sim 0.3-0.7c$
- (c) No interior echo: waves that entered the null zone do NOT re-emerge
(null zone blocks EM propagation; doc_orbit_pressure Section 3)
- (d) Shell pressure signature: enhanced X-ray/UV emission at the jet base
from maximum-density cell compression at the boundary

OPEN: compute the shell reflection geometry quantitatively.
Reference: Cendes et al. 2022, ApJ 924, 65 (AT2018hyz)

1.2 Long-lived atmospheric vortices and torsion medium stabilization

PREDICTION: Rotating atmospheric systems that persist beyond the torsion medium's characteristic relaxation timescale can drive quasi-static cell depletion at the vortex axis. The resulting depleted axis acts as a persistent low-pressure node that self-reinforces the rotation.

CANDIDATE: JUPITER'S GREAT RED SPOT

Duration: ~ 350 years -- far exceeds the relaxation threshold.
Wind speed at edge: ~ 150 m/s. Acoustic intensity at the rotating wall far exceeds the medium's saturation threshold.
PREDICTION: The Great Red Spot axis has a partially depleted torsion medium zone that self-reinforces the vortex's longevity.
Falsifiable: compare vortex stability vs persistence time across Jovian storms. Long-lived storms should show anomalous gravity gradient at the axis (Juno spacecraft gravity data is available for this test).

EARTH TORNADOES: duration minutes to hours, far below the relaxation threshold. Purely elastic medium response; no quasi-static depletion.

TRANSIENT NATURAL OBSERVATIONS (elastic regime, sub-threshold duration):
Ball lightning: a lightning strike creates a brief high-intensity rotating field. Torsionverse: a transient depleted zone forms at the vortex axis with a hyper-compressed boundary shell. The glow IS the boundary's pressure being relieved -- the compressed shell propagates T_{1g} (EM pressure) waves outward as it equilibrates. EM and pressure are the same thing in this framework; the light is the pressure. Passes through walls because the interior has no cells to interact with. Collapses with a pop when the driving field stops: the stored pressure releases as a single pressure wave.

Sonoluminescence: acoustic collapse creates a hyper-compressed shell at the bubble wall. The light flash IS the pressure wave from that shell propagating outward as it relieves. Duration ~ 100 ps = the propagation time through the compressed region. Temperature $\sim 10,000$ K = the energy density of the compressed cell boundary converted to thermal as cells relax.

These natural transient cases confirm the mechanism qualitatively.

STORM FORMATION FROM PLANETARY WAKES:

A rotating body's Zone 3 co-rotation sets nearby cells into partial co-rotation as it passes. When the body moves on, those cells are still circulating -- in a shear vortex state (T_{2g} excitation) -- and have not yet returned to equilibrium. The wake is a region of residual shear circulation, not depleted medium. The medium has no preferred shape; it only equalizes pressure and shear. The circulating excitation dissipates over the medium's relaxation timescale.
When Earth enters this circulating wake, the cells around Earth are already in partial co-rotation from the passing body's Zone 3 influence. This modifies the pressure landscape that Earth's own Zone 3 field sits within, creating atmospheric pressure anomalies that manifest as weather perturbations.
Prediction: correlation between Earth crossing Jupiter's orbital wake and large storm frequency on Earth. The relevant lag = Jupiter's orbital period minus Earth's response time. The circulation handedness (co-rotating same or opposite direction as Earth's rotation) determines whether the interaction produces a high or low pressure anomaly.

EARTH POLAR VORTEX: marginal case when multi-year. OPEN: quantify.

2. NUCLEAR APPLICATIONS

2.1 Muon-catalysed fusion anti-sticking mechanism

[doc_hadron_manipulation Section 6.3 -- specific, falsifiable prediction]
Required: 20.90 MeV circularly polarized photons at He-4 Zone 2 in MCF plasma.
Derivation: $E_{\text{vortex}} = R_s \cdot \hbar c / r_{\text{He4}} = R_s \cdot m_{\pi} c^2 = 20.90 \text{ MeV}$ (exact).
Prediction: sticking coefficient ω_s reduced; >10x more fusions per muon.
Facility: PSI muon beam + 0.94 GeV electrons + Compton laser (~\$10M addition).
Falsifiable: measure ω_s with vs without the vortex field.

2.2 Hg-197 -> Au-197 pressure valley mechanism

[doc_hadron_manipulation Section 5.3 -- retrodict, derivation of known reaction]
Torsionverse explanation: neutron capture Hg-196 -> Hg-197 places the nucleus past the Z-N pressure valley ridge; it slides to Au-197 via the Zone 2/3 pressure landscape. The 64.14-hour half-life set by Zone 3 weak coupling ($\sin^2(\theta_w) = 0.22290$, derived exactly from I_h geometry, $4.6e-6$ from PDG).

3. MOLECULAR APPLICATIONS

3.1 O-H bond disruption: IRMPD (established baseline, fully scripted)

[bond_disruption_em.py 15/15 PASS -- establishes the sequential ladder baseline]
11.2 photons at 2.70 μm (111 THz) required per O-H bond. Fully derived.

3.2 A_g orthogonal-beam bond breaking (new prediction from I_h group theory)

$T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$ [CG exact]. At crossing angle = 72 deg (C5):
 $\chi(T_{1g}, C5) = \phi \rightarrow \phi^2 = 2.618 \times$ more A_g than at 90 deg.
Prediction: bond-breaking cross-section at 72 deg / 90 deg = $\phi^2 = 2.618$.
No existing framework predicts this ratio or its group-theoretic origin.
Falsifiable: two OPO lasers at 2.70 μm , crossed at 72 vs 90 vs other angles.

3.3 Magnetic pre-alignment enhancement

Water dipole moment = 1.85 Debye. Pre-aligning molecules with a field pulse before the excitation laser converts statistical to coherent excitation, multiplying the ϕ^2 effect. Pulse sequence: align (ns) -> excite (fs).

3.4 Water splitting application

$\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{O}_2$ via A_g direct excitation at 111 THz, 72 deg crossing.
 $\Delta G = 237 \text{ kJ/mol} = 2.46 \text{ eV/molecule}$. The A_g mechanism bypasses the sequential IRMPD ladder; potential threshold reduction by factor ϕ^2 .

4. COSMOLOGICAL APPLICATIONS

4.1 Tidal disruption event jet delay: AT2018hyz

[See Section 1.1 -- $T_{2g} \times T_{2g} \rightarrow H_g$ jet emission from hyper-dense boundary]

4.2 Cosmological redshift — candidate mechanisms and void-edge prediction

[doc_redshift.txt: Candidate Mechanisms and Void-Edge Prediction]

Candidate C (cell migration): cell density decreases as volume expands.
Observable: dark energy equation of state $w \neq -1$ (DESI 2024 consistent).
JWST prediction: $a_0(z=1) = 2.086e-10 \text{ m/s}^2$ (79% higher than today) [RS6].

Candidate D (PRIMARY NEW RESULT): sparse void edges create lower local medium frequency $f_{\text{cell}} = c/L_J^{\text{local}}$. Redshift arises at void-edge density gradients:
 $z + 1 = (n_{\text{filament}} / n_{\text{void_edge}})^{1/3}$ [zero free parameters, RS12]
KBC void (25% local underdensity): H_0 correction ~9% in correct direction.

Transfer picture: $\lambda_{\text{visible}} / L_J \sim 5 \times 10^{10}$ -- wave sees continuous medium;
no per-cell interaction; mechanism is a collective medium property.

HELIOSPHERE / HUBBLE CONSISTENCY CHECK [RS14]: using the OBSERVED Hubble rate
($H_0 = 67.4$ km/s/Mpc, imported from Planck 2018 -- not derived from torsionverse),
cells migrating outward over 13.8 Gyr travel 113.9 AU. The heliopause is at 120 AU.
Time to reach 120 AU at $H_0 = 14.5$ Gyr (universe age: 13.8 Gyr).
This is a dimensional consistency check using the imported H_0 ; not a TV prediction.

TV-NATIVE AGE FINDING [RS3b-c, cell_expansion_age.py]:

Using $n_{\text{cells}} = 1/L_J^3$ (TV-native, no ρ_{Lambda}) and observed Fermi-LAT gamma
flux at $m_{\text{crit}} = 9.933$ GeV:

$H_{\text{TV}} = 3.3 \times 10^{13} \text{ /s}$ (TV-native expansion rate from cloning alone)

The TV-native timescale for cell expansion to reach the heliosphere (120 AU) is
many orders of magnitude larger than the mainstream 13.8 Gyr. Our cell island
is likely significantly older than the mainstream estimate. The termination shock
at $\sim 86 \text{ AU} = 120/e^{(1/3)} \text{ AU}$ is the self-consistent starting radius [RS3e].

NOTE: exact age not asserted without more supporting data – depends on whether
local galactic or cosmological gamma flux applies, and cloning vs migration
dominance. The robust conclusion is qualitative: significantly older, not 13.8 Gyr.

CELL PROPAGATION HORIZON [RS15]: Hubble radius $c/H_0 \approx 14.5 \text{ Gly}$ is the boundary
beyond which no cells have propagated since the initial high-temperature epoch
($T > E_{\text{cell}}/k_B \sim 10^{15} \text{ K}$, $t < 10^{-8} \text{ s}$ after formation). No cells = no medium =
no Hopf windings = no structured matter. This is a hard physical substrate limit,
not a GR coordinate singularity. Fermi paradox corollary: no civilisation can
cross this boundary as structured matter, regardless of technology.

4.3 Cell cloning condition: $\hbar c/(2L_J) = m_{\text{crit}} = 9.933 \text{ GeV}$ (exact)

Algebraic identity connecting Bragg lattice resonance to winding critical mass.
Same energy as $N_J = 2.000 \text{ EW/sub-cell boundary}$ [PG15 PASS].

5. ACOUSTIC AND HUMAN-SCALE IMPLICATIONS

Full derivations in docs/doc_acoustics.txt (v1.0); chi calculations in
internal_only/scripts/geo_search.py (13/13 PASS, GS1-GS12).

5.1 Chladni patterns: retrodiction via dwell-time asymmetry

Sand grains on a resonating plate migrate to pressure MINIMA (nodal lines)
by dwell-time asymmetry -- the same mechanism that produces gravity. Near a
pressure node the wave spends more time per unit displacement; grains accumulate
there. This is the torsionverse pressure-following law at acoustic frequency.

PREDICTION: Two acoustic sources crossed at 72 deg (C5 icosahedral angle)
produce $\phi^2 = 2.618 \times$ more A_g mode excitation at the crossing than at 90 deg.
Chladni patterns should show ϕ^2 stronger grain concentration at the crossing
point for 72 deg vs 90 deg geometry. Falsifiable: two speakers, adjustable bracket.

5.2 Acoustic levitation: retrodiction via T_{1g} pressure minimum stability

Objects levitated by acoustic standing waves sit at pressure nodes. Standard
explanation: radiation pressure. Torsionverse: objects drift to T_{1g} pressure
minima by dwell-time asymmetry; the node is a stable equilibrium (zero gradient)
for the same reason a planetary orbit is stable.

PREDICTION: A 72-deg crossing geometry levitates objects with $\phi^2 = 2.618 \times$
less required acoustic power than a 90-deg crossing at the same frequency.

5.3 N-ring static coupling: derivation from I_h character table

A ring of N elements with adjacent angular spacing $\theta = 360/N$ has T_{1g} coupling at its center given by $\chi = 1 + 2\cos(\theta)$.

N=3: $\chi = 0$ (exact null; equilateral triangle arrangement)
N=5: $\chi = \phi = 1.618$ (C5 in I_h; maximum)
N=10: $\chi = \phi^2 = 2.618$ (double C5 spacing; non-I_h but stronger)
N→∞ (circle): $\chi \rightarrow 3$ (full sphere; maximum possible)

IMPORTANT: N=4 RING (adjacent 90-deg spacing): $\chi = +1$ (positive).
4-sided PYRAMID (opposite face normals, 180-deg): $\chi = -1$.
Same count, different dominant angle, opposite sign.

5.4 Stone circles as static carousels

A ring of N upright standing stones acts as the static version of a spinning N-pyramid carousel, driven by seismic background rather than centrifugal force. N-dependence is identical to the dynamic case.

STONEHENGE inner horseshoe: 5 trilithons with approximate C5 symmetry.
 $\chi(T_{1g}, C5) = \phi$ at the altar stone (the focal point of the horseshoe).
If the arrangement was deliberate, this is the optimal ring geometry from I_h character theory.

PREDICTION: Stone circles with N=5 (or multiples of 5) show stronger documented acoustic phenomena at their centers than N=3 or N=4 circles.

5.5 Ringing rocks: internal crystal resonance reduces apparent weight

Rocks that produce a clear musical tone when struck (Pennsylvania Ringing Rocks Park, Stonehenge bluestones) have olivine/pyroxene inclusions with approximate icosahedral packing in their atomic coordination shells. Seismic background continuously drives these rocks at their natural resonance. At resonance: internal standing wave creates a T_{1g} pressure pattern; pressure minimum inside the rock → the rock is lighter than a non-resonant rock of the same material and size.

PREDICTION: Ringing rocks are measurably lighter per unit volume than non-ringing rocks of the same parent material and size, with weight deficit proportional to the resonance quality factor Q. Effect reverses when driving stops (dynamic eddy; not a permanent T_{2g} depletion).

5.6 Newgrange chamber: resonance calculation

Newgrange, Ireland (c. 3200 BCE). Main chamber R ~ 3 m, passage L ~ 19 m.
Chamber spherical fundamental: $f = v/2R = 57$ Hz (published archaeoacoustics: ~110 Hz; actual corbelled geometry has R_{eff} ~ 1.56 m → $f = 110$ Hz).
Passage Helmholtz resonance (passage = neck, chamber = cavity): $f = 1.18$ Hz (infrasound).
Both modes create T_{1g} pressure minima at the chamber center via seismic driving.

6. GYROSCOPE PRECESSION AND FLUID VORTEX SENSING

Full derivations in docs/doc_gyroscope.txt (10/10 PASS, GY1-GY10).

6.1 Bernoulli mechanism for precession: retrodiction

A spinning gyroscope rim drags the torsion medium at the equator. By Bernoulli:

lower pressure at the equatorial rim; higher pressure at the poles. When a torque tips the spin axis, part of the low-pressure equatorial belt moves into a higher-pressure region. The pressure-gradient force is a restoring torque perpendicular to both spin and applied torque -> precession, not further tipping.

At lab scale: torsion medium density is negligible vs air drag. The mechanism is correct but the medium Bernoulli term is unobservable at lab scale; standard mechanics applies. This is a retrodiction: same result, new physical origin.

At Zone 2 scale (molecular, nuclear, neutron star): medium pressure IS the dominant mechanism. NMR precession and pulsar spin-down are Zone 2 phenomena.

6.2 Two-axis fluid gyroscope: zero cross-coupling (exact)

For a 2-axis fluid gyroscope (fluid circulating in orthogonal rings), the

T_{1g} modes along the two axes decouple exactly:

$$\chi(T_{1g_x} * T_{1g_y}, C_2) = (-1) * (-1) = +1 = \chi(A_g)$$

A_g is the totally-symmetric I_h representation: zero cross-coupling is exact,

not approximate. This is a new structural result: the decoupling follows from group theory, not from approximate symmetry or engineering tolerance.

PREDICTION: A fluid gyroscope built with exact C₅ symmetry in its ring geometry will show ϕ^2 less cross-axis coupling than a C₄ (rectangular) geometry. Measurable in precision inertial navigation units.

6.3 Critical size R_{crit} and pulsar timing glitches

The medium shear wave speed $v_s = R_s * c$ limits how quickly the medium can redistribute Bernoulli pressure from a spinning object. For objects larger than:

$$R_{crit} = v_s * \tau_{relax} \approx 355 \text{ AU}$$

the medium takes longer than τ_{relax} to equilibrate -> gyroscopic precession rates become time-varying on the τ_{relax} timescale.

PULSAR TIMING GLITCHES: pulsars are rotating neutron stars with Zone 2 as the dominant medium zone. The glitch timescale (sudden spinup then exponential relaxation over weeks to months) matches the Zone 2 τ_{relax} timescale.

PREDICTION: glitch recovery timescale $\tau_{recovery} \sim \tau_{relax}(\text{Zone } 2)$.
Falsifiable: compile glitch recovery times across known glitching pulsars; compare distribution to the Zone 2 τ_{relax} prediction.

7. SPINNING PYRAMID AND ELECTRON ESCALATOR

Full derivations in docs/series2/doc_series2_spinning_pyramid.txt and docs/series2/doc_series2_piezo_escalator.txt;

chi calculations in analysis/demos/acoustics_doc.py (27/27 PASS, AC1-AC15, SP1-SP7c).

7.1 Electron escalator: passive free-electron aggregator

The pentagonal pyramid (face slope $\arctan(1/\phi) = 31.72^\circ$, $\chi(T_{1g}, C_5) = \phi$) has its face cells permanently at the Maxwell-critical icosahedral orientation. This is the SAME mechanism as piezoelectricity: a piezo crystal requires mechanical stress to deform cells to the critical angle; the pyramid geometry is permanently at that angle with no applied stress. Free electrons from the surrounding environment are guided toward the apex by the T_{1g} χ pressure gradient with no moving parts.
Mode 1 (isolated): accumulates to Coulomb saturation in < 1 second; passively

maintained against leakage indefinitely.

Mode 2 (base wire to ground): continuous DC current, rate set by $\eta \times \chi$.

COOLING PREDICTION: each collected electron's thermal KE (~ 0.038 eV at 300K) is removed from the surrounding air -- passive local cooling at the collection site.

7.2 Spinning pyramid: two propulsion modes

$\chi(T_{1g}, C5)$ = ϕ under rotation produces two mechanically distinct effects:

Mode 1 (THRUSTER): escalator F1 + propeller F2 deliver a felt forward push.

Mode 2 (LOW PRESSURE): T_{1g} χ coupling sustains low-pressure column on

spin axis; apparent weight of objects on axis decreases proportional to ω^2 .

Key discriminator: 4-sided pyramid INCREASES weight ($\chi = -1$); 3-sided is null ($\chi = 0$). Conventional aerodynamics predicts the same sign for all -- the sign

reversal between 4-sided and 5-sided falsifies the aero-only explanation.

Testable for under \$30 with a precision scale, hobby motor, and sheet aluminium.

7.3 N=5 thrust ring: satellite station-keeping

N=5 ring of pyramids placed ahead of a craft (apexes pointing rearward creates low-pressure zone at craft nose; craft falls forward):

$dP = B_{\text{eff}} \times \text{dep} \times \chi = 1.68$ Pa at A1 tip-speed limit

Force = $dP \times A = 1.3$ N on 1m-dia 100 kg craft = 1.35 milli-G [SP7c]

χ factor = $\phi^2 + N\phi = 10.71$ (carousel + individual rearward).

Mode 2 simultaneously recovers electrical power: self-powered, no propellant.

Safety [SP5-SP6]: all practical configurations are 9+ orders of magnitude below any pressure-related hazard; structural failure of the pyramid occurs first.

8. COMPANION PAPERS

SERIES 1 (published, Zenodo):

[doc_alpha]	https://doi.org/10.5281/zenodo.22013651
[doc_torsion]	https://doi.org/10.5281/zenodo.22016573
[doc_higgs]	https://doi.org/10.5281/zenodo.22032555
[doc_jobson_cell]	https://doi.org/10.5281/zenodo.22032906
[doc_magnetism]	https://doi.org/10.5281/zenodo.22036406
[doc_nucleus]	https://doi.org/10.5281/zenodo.22042337
[doc_orbit_pressure]	https://doi.org/10.5281/zenodo.22044378
[doc_entanglement]	https://doi.org/10.5281/zenodo.22052350
[doc_qm]	https://doi.org/10.5281/zenodo.22052395
[doc_leptons]	https://doi.org/10.5281/zenodo.22057421
[doc_particle_gen]	https://doi.org/10.5281/zenodo.22068557
[doc_electron]	https://doi.org/10.5281/zenodo.22105622

SERIES 2 (companion papers; v2 adds Sections 5-6 to this synthesis):

doc_hadron_manipulation.txt	-- MCF anti-sticking; Hg→Au mechanism DOI: https://doi.org/10.5281/zenodo.2214
doc_series2_muon_lubrication.txt	-- MCF anti-sticking vortex: standalone prediction (outline) DOI: https://doi.org/10.5281/zenodo.2214
doc_chemistry.txt	-- A _g bond excitation; water splitting DOI: https://doi.org/10.5281/zenodo.2214
doc_redshift.txt	-- Cosmological redshift: candidate mechanisms and void-edge prediction DOI: https://doi.org/10.5281/zenodo.2214
doc_gyroscope.txt	-- Gyroscope precession; fluid vortex sensing DOI: https://doi.org/10.5281/zenodo.2214
doc_acoustics.txt	-- Acoustic implications; ringing rocks; stone circles DOI: https://doi.org/10.5281/zenodo.2214
doc_series2_piezo_escalator.txt	-- Piezo + electron escalator (static T_{1g} coupling, outline) DOI: https://doi.org/10.5281/zenodo.2214
doc_series2_spinning_pyramid.txt	-- T_{1g} χ coupling; apparent weight reduction; satellite thruster ring DOI: https://doi.org/10.5281/zenodo.2214
doc_series2_fluid_bells.txt	-- Counter-rotating fluid bells; Coriolis roll; T_{2g} thrust DOI: https://doi.org/10.5281/zenodo.2214